

drnek. $\int \sin 2x \cos 5x \, dx = ?$

$$= \frac{1}{2} \int [\sin (2+5)x + \sin (2-5)x] \, dx$$

$$= \frac{1}{2} \int \sin 7x \, dx - \frac{1}{2} \int \sin 3x \, dx$$

$$= -\frac{1}{14} \cos 7x + \frac{1}{6} \cos 3x + C$$

7.5 Trigonometrik ve Hiperbolik değişken değiştirme

Trigonometrik dönüşümler, $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, ve $\sqrt{x^2 - a^2}$ içeren integralleri doğrudan hesaplayabildiğimiz integrallere dönüştürmekte etkili olabilir.

$$x = a \tan \theta \text{ ile } (dx = a \sec^2 \theta d\theta)$$

$$a^2 + x^2 = a^2 + a^2 \tan^2 \theta = a^2(1 + \tan^2 \theta) = a^2 \sec^2 \theta.$$

$$x = a \sin \theta \text{ ile } (dx = a \cos \theta d\theta)$$

$$a^2 - x^2 = a^2 - a^2 \sin^2 \theta = a^2(1 - \sin^2 \theta) = a^2 \cos^2 \theta$$

$$x = a \sec \theta \text{ ile } (dx = a \sec \theta \tan \theta d\theta)$$

$$x^2 - a^2 = a^2 \sec^2 \theta - a^2 = a^2(\sec^2 \theta - 1) = a^2 \tan^2 \theta.$$

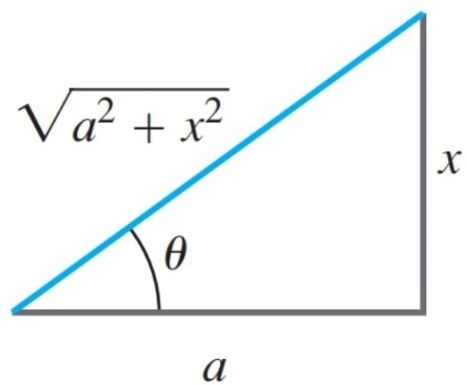
elde edilir.

Bir integrasyonda kullandığımız herhangi bir değişken dönüşümünün, daha sonra esas değişkenlere dönebilmemiz için tersinir olmasını isteriz. Örneğin, $x = a \tan \theta$ ise, integrasyon yapıldıktan sonra $\theta = \tan^{-1}(x/a)$ yazabilmeyi isteriz. $x = a \sin \theta$ ise, işimiz bittikten sonra $\theta = \sin^{-1}(x/a)$ yazabilmeyi ve aynı işlemleri $x = a \sec \theta$ için de yapabilmeyi isteriz.

$$x = a \tan \theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2} \quad \text{olmak üzere} \quad \theta = \tan^{-1}\left(\frac{x}{a}\right) \text{ gerektirir.}$$

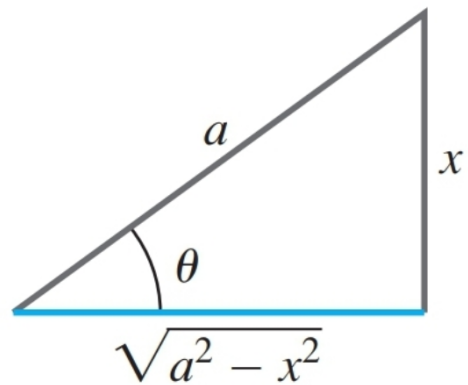
$$x = a \sin \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} \quad \text{olmak üzere} \quad \theta = \sin^{-1}\left(\frac{x}{a}\right) \text{ gerektirir.}$$

$$x = a \sec \theta, \quad \begin{cases} 0 \leq \theta < \frac{\pi}{2} & \frac{x}{a} \geq 1 \text{ ise} \\ \frac{\pi}{2} < \theta \leq \pi & \frac{x}{a} \leq -1 \text{ ise} \end{cases} \quad \text{olmak üzere} \quad \theta = \sec^{-1}\left(\frac{x}{a}\right) \text{ gerektirir.}$$



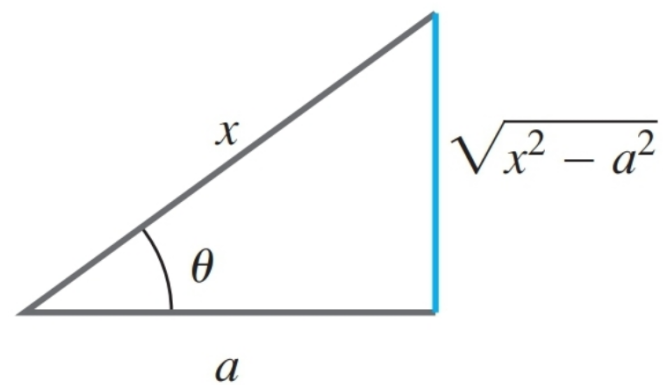
$$x = a \tan \theta$$

$$\sqrt{a^2 + x^2} = a |\sec \theta|$$



$$x = a \sin \theta$$

$$\sqrt{a^2 - x^2} = a |\cos \theta|$$



$$x = a \sec \theta$$

$$\sqrt{x^2 - a^2} = a |\tan \theta|$$

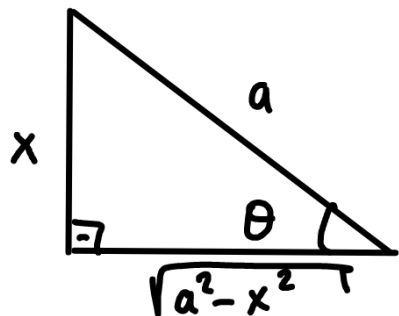
örnek. $\int \sqrt{a^2 - x^2} dx = ? \quad (a > 0)$

$$x = a \sin \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

$$dx = a \cos \theta d\theta$$

$$I = \int \sqrt{a^2 - a^2 \sin^2 \theta} \cdot a \cos \theta d\theta = a^2 \int \cos^2 \theta d\theta$$

$$= \frac{a^2}{2} \int (1 + \cos 2\theta) d\theta = \frac{a^2}{4} \sin 2\theta + \frac{a^2}{2} \theta + C$$



$$= \frac{a^2}{4} 2 \sin \theta \cos \theta + \frac{a^2}{2} \theta + C$$

$$= \frac{a^2}{2} \frac{x}{a} \cdot \frac{\sqrt{a^2 - x^2}}{a} + \frac{a^2}{2} \arcsin \frac{x}{a} + C$$

Hiperbolik değişken değiştirmeler :

$\sqrt{x^2 + a^2}$ ve $\sqrt{x^2 - a^2}$ gibi ifadeler için :

$$x = a \sinh u ; \quad -\infty < u < \infty$$

$$dx = a \cosh u du \quad \Rightarrow \quad \sqrt{x^2 + a^2} = \sqrt{a^2 \sinh^2 u + a^2} = a \cosh u$$

$$x = a \cosh u ; \quad 0 \leq u < \infty$$

$$dx = a \sinh u du \quad \Rightarrow \quad \sqrt{x^2 - a^2} = \sqrt{a^2 \cosh^2 u - a^2} = a \sinh u$$

Prnek. $\int \sqrt{x^2 + a^2} \, dx, \, a > 0$

$$x = a \sinh u \Rightarrow dx = (\cosh u \, du) \cdot a$$

$$I = \int \sqrt{x^2 + a^2} \, dx = a^2 \int \cosh^2 u \, du = \frac{1}{2} \int (\cosh 2u + 1) \, du$$

$$= \frac{a^2}{4} \sinh 2u + \frac{a^2}{2} u + C$$

$$\int \sqrt{x^2 + a^2} \, dx = \frac{1}{2} a^2 \sinh u \cosh u + \frac{1}{2} a^2 u + C$$

$$\sinh u = \frac{x}{a} \Rightarrow u = \sinh^{-1} \frac{x}{a}, \quad \cosh u = \sqrt{1 + \sinh^2 u} = \frac{\sqrt{x^2 + a^2}}{a}$$

$$\int \sqrt{x^2 + a^2} dx = \frac{1}{2} x \sqrt{x^2 + a^2} + \frac{1}{2} a^2 \sinh^{-1} \frac{x}{a} + C$$

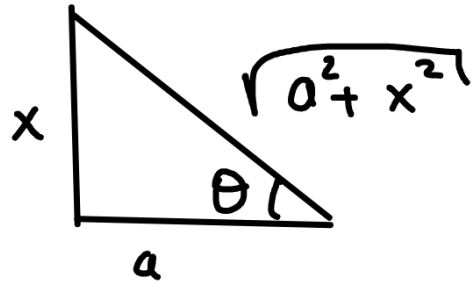
Aynı sonucu trigonometrik değişken kullanarak yapalım:

$$\int \sqrt{x^2 + a^2} dx = ?$$

$$x = a \tan \theta \Rightarrow dx = a \sec^2 \theta d\theta$$

$$I = \int \sqrt{a^2 \tan^2 \theta + a^2} a \sec^2 \theta d\theta = a^2 \int \sec^3 \theta d\theta$$

$$= \frac{1}{2} a^2 \sec \theta \tan \theta + \frac{1}{2} a^2 \ln |\sec \theta + \tan \theta| + C$$



$$I = \frac{1}{2} a^2 \frac{\sqrt{a^2 + x^2}}{a} \cdot \frac{x}{a} + \frac{1}{2} a^2 \ln \left| \frac{x + \sqrt{x^2 + a^2}}{a} \right| + C$$

$$I = \frac{1}{2} x \sqrt{a^2 + x^2} + \frac{1}{2} a^2 \ln (x + \sqrt{a^2 + x^2}) + C$$

↓
 $(-\frac{a^2}{2} \ln a)$ ya C 'ye katabiliriz)

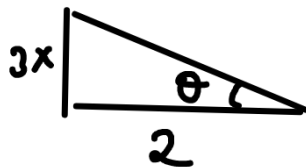
örnek. $\int \frac{dx}{(9x^2 + 4)^{3/2}} = ?$

$$x = \frac{2}{3} \tan \theta \Rightarrow dx = \frac{2}{3} \sec^2 \theta d\theta$$

$$I = \int \frac{2}{3} \sec^2 \theta \frac{1}{\left(9 \frac{4}{9} \tan^2 \theta + 4\right)^{3/2}} d\theta$$

$$= \int \frac{2}{3} \sec^2 \theta \frac{1}{8 \cdot \sec^3 \theta} d\theta = \frac{1}{12} \int \cos \theta d\theta$$

$$= \frac{1}{12} \sin \theta + C$$



$$I = \frac{1}{12} \frac{3x}{\sqrt{9x^2 + 4}} + C$$

drnek. $\int \frac{dx}{(16x^2-25)^{3/2}} = ?$

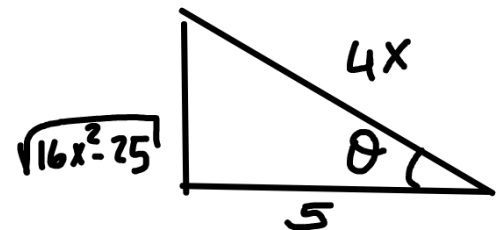
$$x = \frac{5}{4} \sec \theta; \quad (0 < \theta < \frac{\pi}{2})$$

$$dx = \frac{5}{4} \sec \theta \tan \theta d\theta$$

$$(16x^2-25)^{3/2} = (25 \sec^2 \theta - 25)^{3/2} = 125 \tan^3 \theta$$

$$I = \int \frac{5}{4} \sec \theta \tan \theta \frac{1}{125 \tan^3 \theta} d\theta = \frac{1}{100} \int \csc \theta \cot \theta d\theta$$

$$= -\frac{1}{100} \csc \theta + C = -\frac{1}{25} \frac{x}{\sqrt{16x^2-25}} + C$$

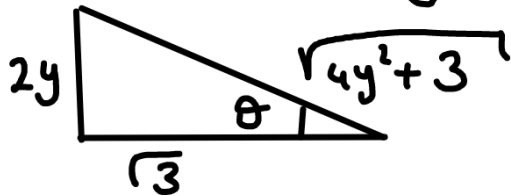


örnek. $\int \frac{dx}{\sqrt{x^2 + x + 1}} = ?$

$$x^2 + x + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} = y^2 + \frac{3}{4} \quad ; \quad y = x + \frac{1}{2}$$

$$I = \int \frac{dy}{\sqrt{y^2 + \frac{3}{4}}} \quad ; \quad y = \frac{\sqrt{3}}{2} \tan \theta \Rightarrow dy = \frac{\sqrt{3}}{2} \sec^2 \theta d\theta$$

$$= \int \frac{\frac{\sqrt{3}}{2} \sec^2 \theta}{\frac{\sqrt{3}}{2} \sec \theta} d\theta = \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + C$$



$$= \ln \left| \frac{\sqrt{4y^2 + 3}}{\sqrt{3}} + \frac{2y}{\sqrt{3}} \right| + C$$

$$= \ln(\sqrt{4y^2+3} + 2y) + C$$

$$= \ln\left[\sqrt{4\left(x+\frac{1}{2}\right)^2+3} + 2\left(x+\frac{1}{2}\right)\right] + C$$

$$= \ln\left(\sqrt{x^2+x+1} + x + \frac{1}{2}\right) + C$$

örnek. $\int_1^3 \frac{x}{\sqrt{4x-x^2}} dx = ?$

$$4x - x^2 = 4 - (x-2)^2 = 4 - y^2 \quad ; \quad y = x-2, \quad dx = dy$$

$$I = \int_{-1}^1 \frac{2+y}{\sqrt{4-y^2}} dy = \underbrace{2 \int_{-1}^1 \frac{dy}{\sqrt{4-y^2}}}_{y=2\sin\theta} + \underbrace{\int_{-1}^1 \frac{y}{\sqrt{4-y^2}} dy}_{=0 \text{ (tek fonks.)}}$$

$$= \arcsin \frac{y}{2} \Big|_{-1}^1 = \frac{2\pi}{3}$$

örnek. $\int \frac{1 + \cot x}{1 + e^{-x} \csc x} dx$ belirsiz integralini bulunuz.

$$\frac{1 + \cot x}{1 + e^{-x} \csc x} = 1 + \frac{\cos x}{\sin x} \bigg/ 1 + \frac{1}{e^x} \cdot \frac{1}{\sin x}$$

$$= \frac{\sin x + \cos x}{\cancel{\sin x}} \cdot \frac{\cancel{e^x \sin x}}{1 + e^x \sin x} = \frac{e^x \sin x + e^x \cos x}{1 + e^x \sin x}$$

$$I = \int \frac{e^x \sin x + e^x \cos x}{1 + e^x \sin x} dx ,$$

$1 + e^x \sin x \rightarrow f(x) = 1 + e^x \sin x$

$$f'(x) = e^x \sin x + e^x \cos x$$

$$\begin{aligned} I &= \int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C \\ &= \ln |1 + e^x \sin x| + C \end{aligned}$$

Örnek. $\int \frac{(x^2-1)^{3/2}}{x} dx$

$$x = \sec \theta \Rightarrow dx = \sec \theta \tan \theta d\theta$$

$$I = \int \frac{(x^2-1)^{3/2}}{x} dx = \int \frac{(\sec^2 \theta - 1)^{3/2}}{\sec \theta} \cdot \sec \theta \tan \theta d\theta$$

$$= \int \frac{(\tan^2 \theta)^{3/2}}{\sec \theta} \sec \theta \tan \theta d\theta$$

$$= \int \tan^4 \theta d\theta$$

$$= \int \tan^2 \theta \tan^2 \theta d\theta$$

$$= \int \tan^2 \theta (\sec^2 \theta - 1) d\theta$$

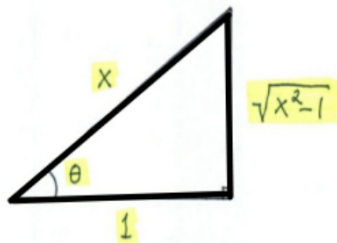
$$= \int \tan^2 \theta \sec^2 \theta d\theta - \int (\sec^2 \theta - 1) d\theta$$

$$= \int \tan^2 \theta \sec^2 \theta d\theta - \int \sec^2 \theta d\theta + \int d\theta$$



$$\tan \theta = u \Rightarrow \sec^2 \theta d\theta = du$$

$$= \frac{u^3}{3} - \tan \theta + \theta + C = \frac{(\tan \theta)^3}{3} - \tan \theta + \theta + C$$



$$\sec \theta = x, \quad \tan \theta = \sqrt{x^2 - 1}$$

$$I = \frac{1}{3} (\sqrt{x^2 - 1})^3 - \sqrt{x^2 - 1} + \operatorname{arcsec} x + C$$

Örnek. $\int \frac{\sqrt{x^2+1}}{x} dx$

$$x = \tan \theta, \quad dx = \sec^2 \theta d\theta$$

$$I = \int \frac{\sqrt{\sec^2 \theta}}{\tan \theta} \sec^2 \theta d\theta$$

$$I = \int \frac{\sec^3 \theta}{\tan \theta} d\theta = \int \frac{\sec \theta \cdot \sec^2 \theta}{\tan \theta} d\theta$$

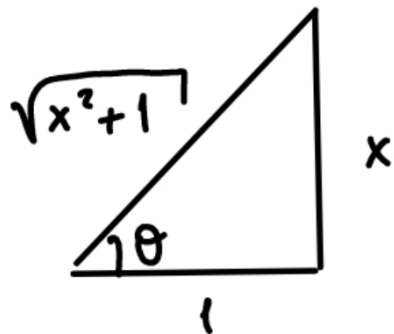
$$= \int \frac{\sec \theta (1 + \tan^2 \theta)}{\tan \theta} d\theta$$

$$= \int \frac{\cos \theta}{\sin \theta} \cdot \frac{1}{\cos \theta} (1 + \tan^2 \theta) d\theta$$

$$= \int \left(\frac{1}{\sin \theta} + \frac{1}{\cos \theta} \frac{\sin \theta}{\cos \theta} \right) d\theta$$

$$= \int \csc \theta d\theta + \int (\sec \theta \cdot \tan \theta) d\theta$$

$$= \ln |\csc \theta - \cot \theta| + \sec \theta + C$$



$$= \ln \left| \frac{\sqrt{x^2 + 1}}{x} - \frac{1}{x} \right| + \sqrt{x^2 + 1} + C$$

Örnek. $\int x^3 \sqrt{4-9x^2} dx$

$$= \int x^3 3 \sqrt{\frac{4}{9} - x^2} dx = 3 \int x^3 \sqrt{\left(\frac{2}{3}\right)^2 - x^2} dx$$

$$x = \frac{2}{3} \sin \theta \Rightarrow dx = \frac{2}{3} \cos \theta d\theta$$

$$= 3 \cdot \int \frac{8}{27} \sin^3 \theta \sqrt{\frac{4}{9} - \frac{4}{9} \sin^2 \theta} \cdot \frac{2}{3} \cos \theta d\theta$$

$$= \int \frac{8}{9} \sin^3 \theta \cdot \frac{2}{3} \cdot \cos \theta \cdot \frac{2}{3} \cdot \cos \theta d\theta$$

$$= \frac{32}{81} \int \sin^3 \theta \cos^2 \theta d\theta$$

$$\cos \theta = u \Rightarrow -\sin \theta d\theta = du$$

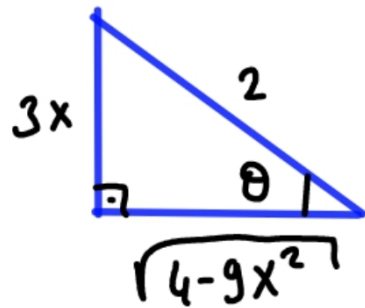
$$= \frac{32}{81} \int (1 - \cos^2 \theta) \cos^2 \theta \cdot \sin \theta d\theta$$

$$= \frac{32}{81} \int (1 - u^2) \cdot u^2 (-du)$$

$$= \frac{32}{81} \int (u^4 - u^2) du$$

$$= \frac{32}{81} \left[\frac{u^5}{5} - \frac{u^3}{3} \right] + C = \frac{32}{81} \left(\frac{1}{5} \cos^5 \theta - \frac{1}{3} \cos^3 \theta \right) + C$$

$$= \frac{32}{81} \left[\frac{u^5}{5} - \frac{u^3}{3} \right] + C = \frac{32}{81} \left(\frac{1}{5} \cos^5 \theta - \frac{1}{3} \cos^3 \theta \right) + C$$



$$\sin \theta = \frac{3x}{2}$$

$$\cos \theta = \frac{\sqrt{4-9x^2}}{2}$$

$$= \frac{4}{243} (4-9x^2)^{3/2} + \frac{1}{405} (4-9x^2)^{5/2} + C$$

Örnek.

$$\int \frac{dx}{(4x^2+4x+5)^2}$$

$$(4x^2+4x+5)^2 = ((2x+1)^2+4)^2 = (y^2+4)^2$$

↓

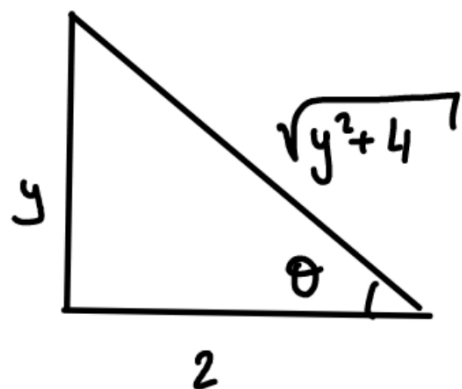
$$y = 2x+1, \quad dy = 2dx$$

$$I = \frac{1}{2} \int \frac{dy}{(y^2+4)^2} ; \quad y = 2\tan\theta ; \quad dy = 2\sec^2\theta d\theta$$

$$I = \frac{1}{2} \int \frac{2\sec^2\theta d\theta}{16(1+\tan^2\theta)^2} = \frac{1}{16} \int \frac{\sec^2\theta}{\sec^4\theta} d\theta$$

$$= \frac{1}{16} \int \cos^2 \theta \, d\theta \quad ; \quad \cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$I = \frac{1}{32} \int (\cos 2\theta + 1) \, d\theta = \frac{1}{64} \sin 2\theta + \frac{\theta}{32} + C$$



$$I = \frac{1}{32} \frac{y}{\sqrt{y^2 + 4}} \cdot \frac{2}{\sqrt{y^2 + 4}} + \frac{1}{32} \arctan \frac{y}{2} + C$$

$$12. \int \frac{dx}{(4x-x^2)^{3/2}}$$

$$\begin{aligned} 4x - x^2 &= - (x^2 - 4x) = - (x^2 - 4x + 4) + 4 \\ &= - (x-2)^2 + 4 \end{aligned}$$

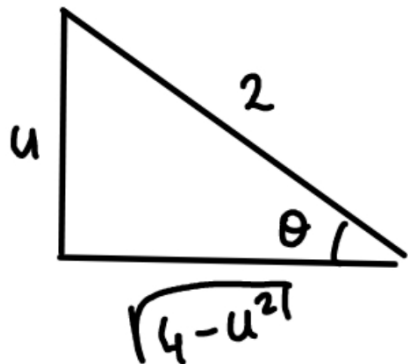
$$\int \frac{dx}{(4x-x^2)^{3/2}} = \int \frac{dx}{(4-(x-2)^2)^{3/2}} \quad ; \quad \begin{aligned} x-2 &= u \\ dx &= du \end{aligned}$$

$$= \int \frac{du}{(4-u^2)^{3/2}} \quad ; \quad u = 2\sin\theta, \quad du = 2\cos\theta d\theta$$

$$= \int \frac{2 \cos \theta \, d\theta}{(4 - 4 \sin^2 \theta)^{3/2}} = \int \frac{2 \cos \theta \, d\theta}{8 \cos^3 \theta} = \frac{1}{4} \int \sec^2 \theta \, d\theta$$

$$= \frac{1}{4} \tan \theta + C = \frac{1}{4} \frac{u}{\sqrt{4-u^2}} + C$$

$$= \frac{1}{4} \frac{(x-2)}{\sqrt{4x-x^2}} + C$$



Örnek.

Verilen indirgeme formüllerini doğrulayınız.

$$1. \quad \int \tan^n x \, dx = \frac{1}{n-1} \tan^{n-1} x - \int \tan^{n-2} x \, dx$$

$$2. \quad \int \sec^n x \, dx = \frac{1}{n-1} \sec^{n-2} x \tan x + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx$$

Çözüm. 1.

$$\begin{aligned} \int \tan^n x \, dx &= \int \tan^2 x \tan^{n-2} x \, dx \\ &= \int (\sec^2 x - 1) \tan^{n-2} x \, dx \\ &= \underbrace{\int \tan^{n-2} x \sec^2 x \, dx}_{\tan x = u \Rightarrow \sec^2 x \, dx = du} - \int \tan^{n-2} x \, dx \\ &= \int u^{n-2} \, du - \int \tan^{n-2} x \, dx \\ &= \frac{1}{n-1} u^{n-1} - \int \tan^{n-2} x \, dx \\ &= \frac{1}{n-1} \tan^{n-1} x - \int \tan^{n-2} x \, dx. \end{aligned}$$

2. Verilen integral

$$\int \sec^n x \, dx = \int \sec^{n-2} x \sec^2 x \, dx$$

biçiminde düzenlebilir. Bu integrali hesaplamak için kısmi integrasyon uygulayalım.

$$\begin{cases} u = \sec^{n-2} x \Rightarrow du = (n-2) \sec^{n-3} x \sec x \tan x \, dx = (n-2) \sec^{n-2} x \tan x \, dx \\ \sec^2 x \, dx = dv \Rightarrow v = \tan x \end{cases}$$

O halde

$$\begin{aligned} \int \sec^n x \, dx &= uv - \int v \, du \\ &= \sec^{n-2} x \tan x - \int (n-2) \sec^{n-2} x \tan^2 x \, dx \\ &= \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x (\sec^2 x - 1) \, dx \\ &= \sec^{n-2} x \tan x - (n-2) \int \sec^n x \, dx + (n-2) \int \sec^{n-2} x \, dx \end{aligned}$$

elde edilir. Dolayısıyla ifade düzenlenirse,

$$\int \sec^n x \, dx + (n-2) \int \sec^n x \, dx = \sec^{n-2} x \tan x + (n-2) \int \sec^{n-2} x \, dx$$

olup buradan

$$(n - 1) \int \sec^n x \, dx = \sec^{n-2} x \tan x + (n - 2) \int \sec^{n-2} x \, dx$$

bulunur. Sonuç olarak

$$\int \sec^n x \, dx = \frac{1}{n-1} \sec^{n-2} x \tan x + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx$$

eşitliğine ulaşılır.

Bu indirgeme formülünü kullanarak, örneğin, $\int \sec^4 x \, dx$ integralini hesaplayalım.

$$\begin{aligned} \int \sec^4 x \, dx &= \frac{1}{4-1} \sec^{4-2} x \tan x + \frac{4-2}{4-1} \int \sec^{4-2} x \, dx \\ &= \frac{1}{3} \sec^2 x \tan x + \frac{2}{3} \int \sec^2 x \, dx \\ &= \frac{1}{3} \sec^2 x \tan x + \frac{2}{3} \tan x + C. \end{aligned}$$

□

Örnek. Verilen integralleri hesaplayınız.

1. $\int_0^{\pi/6} \sin^3 s \cos^3 s \, ds$

2. $\int \sin^4 x \cos^2 x \, dx$

3. $\int_0^{\pi/4} (\tan s)^{3/2} \sec^4 s \, ds$

Çözüm. 1.

$$\begin{aligned} \int_0^{\pi/6} \sin^3 s \cos^3 s \, ds &= \int_0^{\pi/6} (\sin s \cos s)^3 \, ds = \int_0^{\pi/6} \left(\frac{1}{2} \sin 2s\right)^3 \, ds \\ &= \frac{1}{8} \int_0^{\pi/6} \sin^3 2s \, ds \\ &= \frac{1}{8} \int_0^{\pi/6} \sin^2 2s \sin 2s \, ds \\ &= \frac{1}{8} \int_0^{\pi/6} (1 - \cos^2 2s) \sin 2s \, ds \end{aligned}$$

Elde ettiğimiz integrali hesaplamak için $\cos 2s = t$ dönüşümü uygulayalım.

$$\begin{cases} \cos 2s = t \Rightarrow -2 \sin 2s ds = dt \\ s = 0 \Rightarrow t = 1 \\ s = \pi/6 \Rightarrow t = 1/2 \end{cases}$$

Buradan,

$$\begin{aligned} \int_0^{\pi/6} \sin^3 s \cos^3 s ds &= -\frac{1}{16} \int_1^{1/2} (1 - t^2) dt \\ &= -\frac{1}{16} \left(t - \frac{t^3}{3} \right) \Big|_1^{1/2} \\ &= -\frac{1}{16} \left(\frac{1}{2} - \frac{1}{24} - 1 + \frac{1}{3} \right) \\ &= \frac{5}{384} \end{aligned}$$

elde edilir.

2.

$$\begin{cases} \sin^2 x = \frac{1 - \cos 2x}{2} \\ \cos^2 x = \frac{1 + \cos 2x}{2} \end{cases}$$

eşitliklerini yarım-açı formüllerinden biliyoruz. Bu eşitlikleri kullanarak,

$$\begin{aligned}
 \int \sin^4 x \cos^2 x \, dx &= \int (\sin^2 x)^2 \cos^2 x \, dx \\
 &= \int \left(\frac{1 - \cos 2x}{2} \right)^2 \left(\frac{1 + \cos 2x}{2} \right) dx \\
 &= \frac{1}{8} \int (1 - \cos 2x)^2 (1 + \cos 2x) dx \\
 &= \frac{1}{8} \int (1 - \cos^2 2x)(1 - \cos 2x) dx \\
 &= \frac{1}{8} \int (1 - \cos 2x - \cos^2 2x + \cos^3 2x) dx \\
 &= \frac{1}{8} \int dx - \frac{1}{8} \int \cos 2x dx - \frac{1}{8} \int \underbrace{\cos^2 2x}_{\frac{1 + \cos 4x}{2}} dx + \frac{1}{8} \int \underbrace{\cos^3 2x}_{\cos^2 2x \cos 2x} dx \\
 (\star) &= \frac{x}{8} - \frac{\sin 2x}{16} - \frac{1}{16} \int (1 + \cos 4x) dx + \frac{1}{8} \int (1 - \sin^2 2x) \cos 2x \, dx \\
 &= \frac{x}{8} - \frac{\sin 2x}{16} - \frac{1}{16} \left(x + \frac{\sin 4x}{4} \right) + \frac{1}{16} \left(\sin 2x - \frac{\sin^3 2x}{3} \right) + C
 \end{aligned}$$

elde edilir.

NOT: (★) denkleminde en sondaki integral $\sin 2x = t$ dönüşümü yapılarak bir önceki soruya benzer şekilde hesaplanabilir!

3. Verilen integral

$$\begin{aligned}\int_0^{\pi/4} (\tan s)^{3/2} \sec^4 s \, ds &= \int_0^{\pi/4} (\tan s)^{3/2} \sec^2 s \sec^2 s \, ds \\ &= \int_0^{\pi/4} (\tan s)^{3/2} (\tan^2 s + 1) \sec^2 s \, ds\end{aligned}$$

biçiminde düzenlenebilir. Bu integrali hesaplamak için $\tan s = u$ dönüşümüyle değişken değiştirelim.

$$\begin{cases} \tan s = u \Rightarrow \sec^2 s \, ds = du \\ s = 0 \Rightarrow u = 0 \\ s = \pi/4 \Rightarrow u = 1 \end{cases}$$

Buradan,

$$\begin{aligned}\int_0^{\pi/4} (\tan s)^{3/2} \sec^4 s \, ds &= \int_0^1 u^{3/2} (u^2 + 1) du \\&= \int_0^1 (u^{7/2} + u^{3/2}) du \\&= \left(\frac{2}{9} u^{9/2} + \frac{2}{5} u^{5/2} \right) \Big|_0^1 \\&= \frac{2}{9} + \frac{2}{5} \\&= \frac{28}{45}\end{aligned}$$

Örnek.

Verilen integralleri hesaplayınız.

1. $\int_{1/\sqrt{3}}^1 \frac{1}{t(t^2 + 1)} dt$

2. $\int \frac{dx}{\sqrt{x^2 + 3x - 4}}$

3. $\int \frac{x^2}{(1 + x^2)^2} dx$

4. $\int \frac{dx}{x^2 \sqrt{x^2 - 4}}$

ÇÖZÜM. 1. $t = \tan u$ dönüşümü uygulayalım. Bu durumda

$$\begin{cases} dt = \sec^2 u du \\ t = 1/\sqrt{3} \Rightarrow u = \pi/6 \\ t = 1 \Rightarrow u = \pi/4. \end{cases}$$

O halde verilen integral yeniden düzenlenirse;

$$\int_{1/\sqrt{3}}^1 \frac{1}{t(t^2 + 1)} dt = \int_{\pi/6}^{\pi/4} \frac{\sec^2 u}{\tan u \underbrace{(\tan^2 u + 1)}_{\sec^2 u}} du$$

$$\begin{aligned}
&= \int_{\pi/6}^{\pi/4} \frac{1}{\tan u} du \\
&= \int_{\pi/6}^{\pi/4} \frac{\cos u}{\sin u} du \\
&= \int_{\pi/6}^{\pi/4} \frac{(\sin u)'}{\sin u} du \\
&= \ln |\sin u| \Big|_{\pi/6}^{\pi/4} \\
&= \ln \frac{\sqrt{2}}{2} - \ln \frac{1}{2} = \ln \sqrt{2}.
\end{aligned}$$

elde edilir.

2. Verilen integral; paydada, kökün içindeki ifade tam kareye tamamlanarak düzenlenirse

$$\int \frac{dx}{\sqrt{(x + \frac{3}{2})^2 - \frac{25}{4}}}$$

bulunur. Bu integral,

$$\left\{ \begin{array}{l} x + \frac{3}{2} = u \Rightarrow dx = du \end{array} \right.$$

dönüşümü ile

$$\int \frac{du}{\sqrt{u^2 - (5/2)^2}}$$

biçiminde yazılabilir. Elde ettiğimiz bu integrale $u = \frac{5}{2} \cosh t$ hiperbolik dönüşümü uygulayalım. Bu durumda $du = \frac{5}{2} \sinh t \, dt$ olur. O halde

$$\begin{aligned} \int \frac{du}{\sqrt{u^2 - (5/2)^2}} &= \int \frac{(5/2) \sinh t \, dt}{\sqrt{\left(\frac{5}{2} \cosh t\right)^2 - \left(\frac{5}{2}\right)^2}} \\ &= \int \frac{(5/2) \sinh t \, dt}{(5/2) \sinh t} \\ &= \int dt \\ &= t + C \\ &= \cosh^{-1}(2u/5) + C \\ &= \ln \left(\frac{2}{5}u + \sqrt{\frac{4}{25}u^2 - 1} \right) + C \\ &= \ln \left(\frac{2}{5}\left(x + \frac{3}{2}\right) + \sqrt{\frac{4}{25}\left(x + \frac{3}{2}\right)^2 - 1} \right) + C \\ &= \ln \left(\frac{2}{5}\left(x + \frac{3}{2}\right) + \frac{2}{5} \sqrt{x^2 + 3x - 4} \right) + C, \end{aligned}$$

3. ve 4. ödev.